BY : Riya Sunil Kharade

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Abstract

Electronic waste, commonly referred to as e-waste, has become one of the most pressing environmental challenges of the modern era. With the rapid advancement of technology and the increasing consumption of electronic devices, the generation of e-waste has grown exponentially. E-waste includes discarded electrical and electronic equipment such as mobile phones, computers, televisions, refrigerators, and other household or industrial appliances. These items often contain hazardous materials like lead, mercury, cadmium, and other toxic substances. When improperly disposed of, e-waste can lead to severe environmental pollution, contaminating soil, water, and air, and posing serious health risks to humans, including respiratory problems, neurological disorders, and skin diseases.

The management of e-waste is a complex problem due to the lack of structured systems for collection, recycling, and disposal. Traditional disposal methods are often inefficient, unsafe, and environmentally harmful. In response to these challenges, the E-Waste Management System has been developed as a web-based platform aimed at facilitating the efficient collection, processing, and recycling of electronic waste. The system is designed with three main modules: Customers, Administrators, and Kabadiwalas (authorized recyclers). Customers can easily register, log in, and submit pickup requests for their e-waste by providing details such as type, quantity, and preferred pickup date. Administrators validate these requests to ensure proper handling and assign kabadiwalas for collection. The system tracks each request's status through a well-defined workflow: Pending → Scheduled → Collected, enabling users to monitor their requests in real time.
An additional feature of the system is the reward points mechanism, which motivates users to actively participate in recycling. Users earn points for each successful pickup, which can be redeemed for incentives in the future. The platform also includes an awareness and education section to inform users about the environmental impact of e-waste, safe disposal practices, and the benefits of recycling.

In the long term, this system has the potential to integrate with government initiatives, allowing users to receive rewards such as discounts on electricity, water, or property bills. Designed to be secure, scalable, and user-friendly, the E-Waste Management System offers a comprehensive solution to e-waste challenges, promoting environmental sustainability, responsible recycling behavior, and public engagement. By combining technology, awareness, and incentive mechanisms, the platform contributes to creating a safer and cleaner ecosystem, while encouraging communities to adopt environmentally responsible practices on a large scale.

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1. ntroduction

1.1 Introduction

In the modern era, technological growth and digitalization have significantly increased the production and consumption of electronic devices. This trend has resulted in a rapid rise of electronic waste (e-waste), which includes all discarded electrical and electronic equipment such as mobile phones, laptops, televisions, refrigerators, and other household or industrial appliances. These devices often contain hazardous substances, including lead, mercury, cadmium, and other toxic metals, which, if not disposed of properly, pose serious threats to both human health and the environment. E-waste that is mixed with general waste can contaminate soil and groundwater, release toxic fumes during burning, and accumulate harmful chemicals in the ecosystem. This situation highlights the urgent need for effective and systematic e-waste management practices.

The E-Waste Management System has been developed to address these challenges by providing a structured, reliable, and user-friendly platform for the collection, validation, and recycling of electronic waste. The system is designed with three main modules: Customer, Admin, and Kabadiwala (authorized recycler). Customers can register on the system and submit pickup requests for their e-waste, specifying details such as item type, quantity, and preferred pickup schedule. Admins review and validate these requests to ensure authenticity and accuracy, then assign kabadiwalas to collect the waste. The system tracks the status of each request through well-defined stages—Pending → Scheduled → Collected—allowing users to monitor their submissions in real time.

Additionally, the system includes a reward mechanism that encourages users to participate actively in recycling. Users earn points for each completed pickup, which can later be redeemed for rewards or incentives. An awareness and education section provides users with information about the environmental hazards of e-waste, proper disposal methods, and the benefits of recycling. This not only helps in reducing environmental pollution but also promotes responsible behavior among users.

In the future, the system can integrate with government programs to offer additional incentives, such as discounts on utility bills, property taxes, or other public services. The platform is designed to be scalable, secure, and accessible across devices, making it suitable for households, small businesses, and community initiatives. Overall, the E-Waste Management System offers a comprehensive, practical, and environmentally sustainable solution to the challenges posed by electronic waste today.

1.2 Objectives

The primary objective of the E-Waste Management System is to provide an efficient, organized, and user-friendly solution for managing electronic waste. Specifically, the system aims to address the challenges of improper e-waste disposal, lack of awareness, and inefficient collection methods by introducing structured processes and incentivizing responsible recycling behavior.

1. **User Registration and Secure Login:** Enable customers to create accounts and log into the system securely. Proper authentication ensures that only verified users can submit e-waste pickup requests.

2. **Efficient E-Waste Pickup Requests:** Allow users to submit detailed requests specifying the type of electronic items, quantity, and preferred pickup schedule. This ensures that kabadiwalas can collect the waste efficiently and safely.

3. **Admin Validation:** Provide admins with the capability to review, approve, or reject requests. This ensures authenticity and prevents misuse of the system while maintaining smooth workflow.

4. **Real-Time Request Tracking:** Enable customers to monitor the status of their pickup requests through a clear workflow: Pending → Scheduled → Collected. Real-time updates increase transparency and reliability.

5. **Awareness and Education:** Provide educational resources about the environmental impact of e-waste, safe disposal techniques, and benefits of recycling. This promotes environmental consciousness among users.

6. **Support and Query Handling:** Enable users to submit queries or complaints, which are managed efficiently by the admin. Prompt resolution ensures smooth operation and user satisfaction.

7. **Future Integration with Government Programs:** Plan for future scalability where users can receive incentives like utility bill discounts for regular recycling, further promoting public participation.

Overall, these objectives aim to create a sustainable, transparent, and reliable e-waste management system that benefits users, the community, and the environment. The system encourages responsible recycling, reduces environmental hazards, and provides an organized approach to handling electronic waste efficiently.

1.3 Problem Statement

The rapid growth of technology has significantly increased the production and usage of electronic devices. With frequent upgrades and shorter product lifecycles, large amounts of electronic waste (e-waste) are generated every year. E-waste includes devices such as mobile phones, laptops, televisions, refrigerators, and other household or industrial electronics. Improper disposal of these items has become a major environmental concern. Many users either discard e-waste with general waste or store it at home without proper recycling. Such practices lead to toxic substances leaching into the soil and water, release of hazardous gases when burned, and accumulation of dangerous metals in the ecosystem. This poses severe health risks to humans and animals, including respiratory issues, neurological disorders, and other chronic diseases.

One of the key problems is the lack of structured systems for e-waste collection, validation, and recycling. Currently, most collection methods are manual, inefficient, and unreliable. Users have limited knowledge about where and how to dispose of their electronic items safely. There is also no incentive mechanism in most existing methods to encourage people to participate in recycling activities. As a result, e-waste often ends up in unauthorized recycling centers, landfills, or dumpsites, further harming the environment.

Another challenge is tracking the process of collection and recycling. Without a proper system, users are unaware of whether their e-waste has been collected and processed safely. Administrators also face difficulties in monitoring collection, managing recyclers, and ensuring accountability. The lack of awareness and education among the public adds another layer to the problem, reducing participation in responsible e-waste disposal.

The E-Waste Management System aims to solve these issues by providing a comprehensive, web-based solution. It allows users to submit pickup requests, track the collection status, and earn rewards for participation. Admins can validate requests, assign collection tasks to authorized kabadiwalas, and monitor operations efficiently. The system also includes educational content to raise awareness about the environmental hazards of e-waste and the importance of recycling. By integrating technology, rewards, and awareness, the system addresses both the logistical and behavioral challenges of e-waste management, promoting responsible disposal practices and contributing to environmental sustainability.

2. pecific Requirements

2.1 Functional Requirements

Functional requirements define the specific tasks and operations the system must perform. For the E-Waste Management System, these include:

1. **User Registration and Login**: Users must be able to create accounts with personal information such as name, email, and password. Login must be secure to prevent unauthorized access.

2. **E-Waste Pickup Requests:** Users can submit requests for e-waste collection by filling out a form with details like type of item, quantity, and preferred pickup date.

3. **Admin Validation:** Admins review all requests and approve or reject them to ensure authenticity. Validation prevents misuse and guarantees proper handling of e-waste.

4. **Request Tracking:** Each request is tracked through a workflow: Pending → Scheduled → Collected, enabling users to monitor the progress.

5. **Reward Points System:** Users earn points for each successful collection, incentivizing responsible recycling behavior.

6. **Awareness and Education Module:** Provides information about e-waste hazards, safe disposal practices, and the benefits of recycling.

7. **Support and Query Handling: Users** can submit queries, which are addressed by the admin efficiently.

8. **Future Scalability:** The system is designed to integrate with government programs to provide benefits such as discounts on utility bills for frequent recyclers.

2.2 Non-Functional Requirements

Non-functional requirements define the quality, performance, and operational aspects of the system:

1. **Usability:**
   The system must have an intuitive, user-friendly interface accessible to people with minimal technical knowledge.

2. **Security:**
   User data, including personal information and login credentials, must be protected against unauthorized access.

3. **Performance:**
   The system should process requests and updates quickly without delays.

4. **Reliability:**
   The platform must operate consistently without downtime or system crashes.

5. **Scalability:**
   The system should be capable of handling more users, requests, and future integration with government programs.

6. **Compatibility:**
   The platform must function across multiple devices including desktops, laptops, tablets, and smartphones.

7. **Maintainability:**
   The system should be easy to maintain, update, and debug by developers.

8. **Accessibility:**
   The platform should be accessible to users with varying levels of technical knowledge and comply with standard accessibility guidelines.

3. Technology Stack

The E-Waste Management System uses a combination of frontend, backend, and database technologies to ensure a reliable, secure, and responsive application.

Category	Technology/Tool	Purpose/Description
Frontend	HTML	Structures the web pages and content.
	CSS	Provides styling, layout, colors, and responsiveness.
	JavaScript	Adds interactivity, form validation, dynamic content, and real-time status updates.
Backend	PHP	Handles server-side logic, authentication, database communication, and request processing.
Database	MySQL	Stores user data, e-waste requests, reward points, and admin logs securely.
Development Tools	VS Code	Integrated development environment for coding and project development.
Version Control	GitHub	Manages project versions, tracks changes, and enables collaboration.
Optional Integrations	APIs / Government Portals	Enables future features like government program tie-ins and automated notifications.
Testing/Browser	Chrome / Firefox / Edge	Ensures cross-browser compatibility and checks responsiveness on different devices.

3.1 Tech Stack

4. Methodology

4.1 DFD -level 0,1,2

Level 0

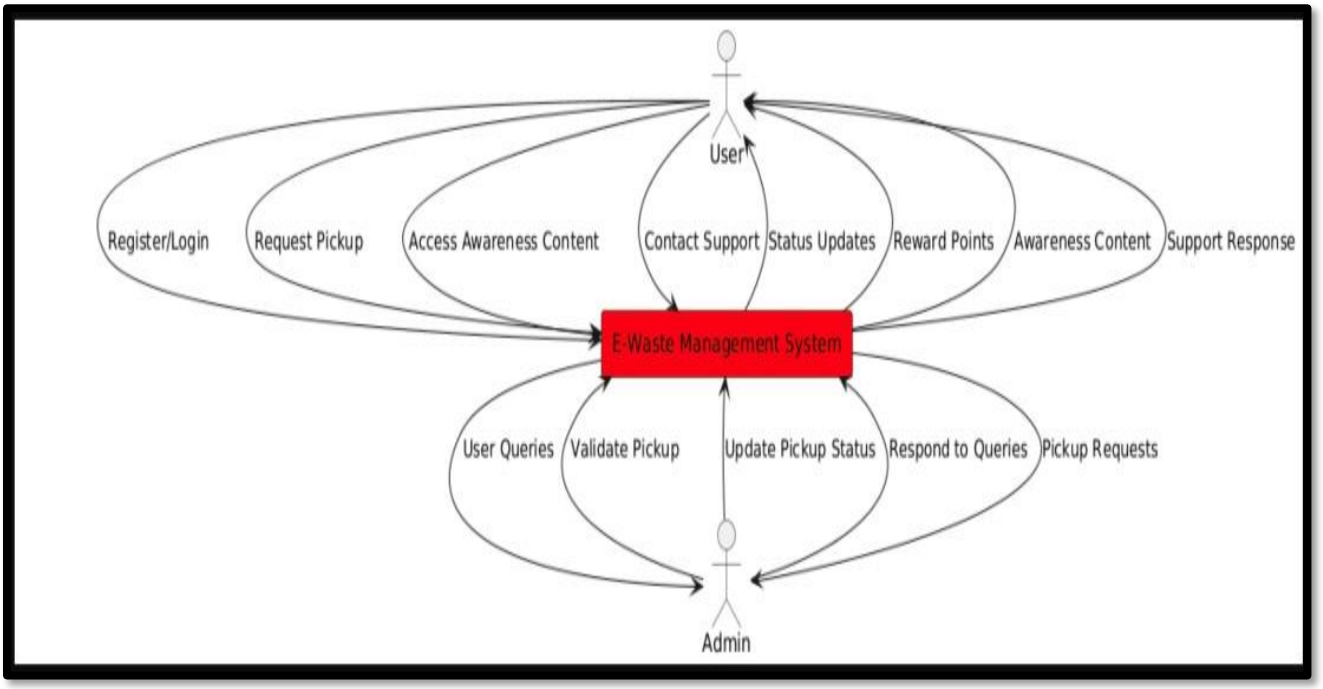


Fig. DFD Level 0

Level 1

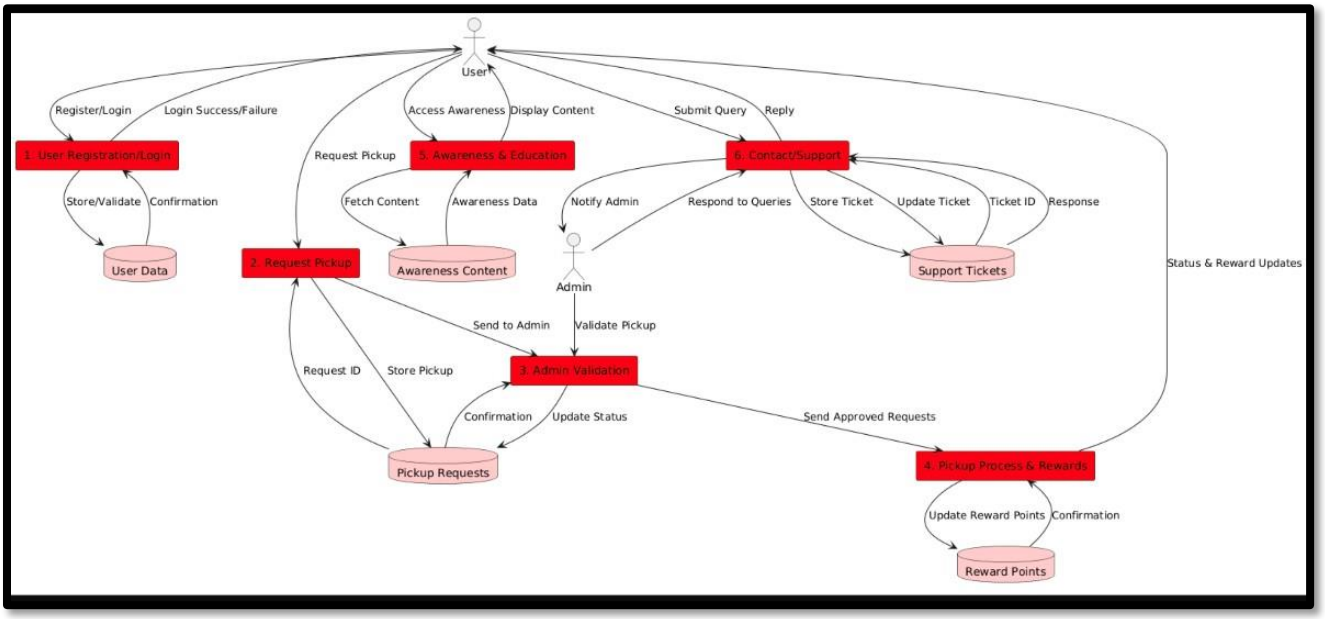


Fig. DFD Level 1

Level 2

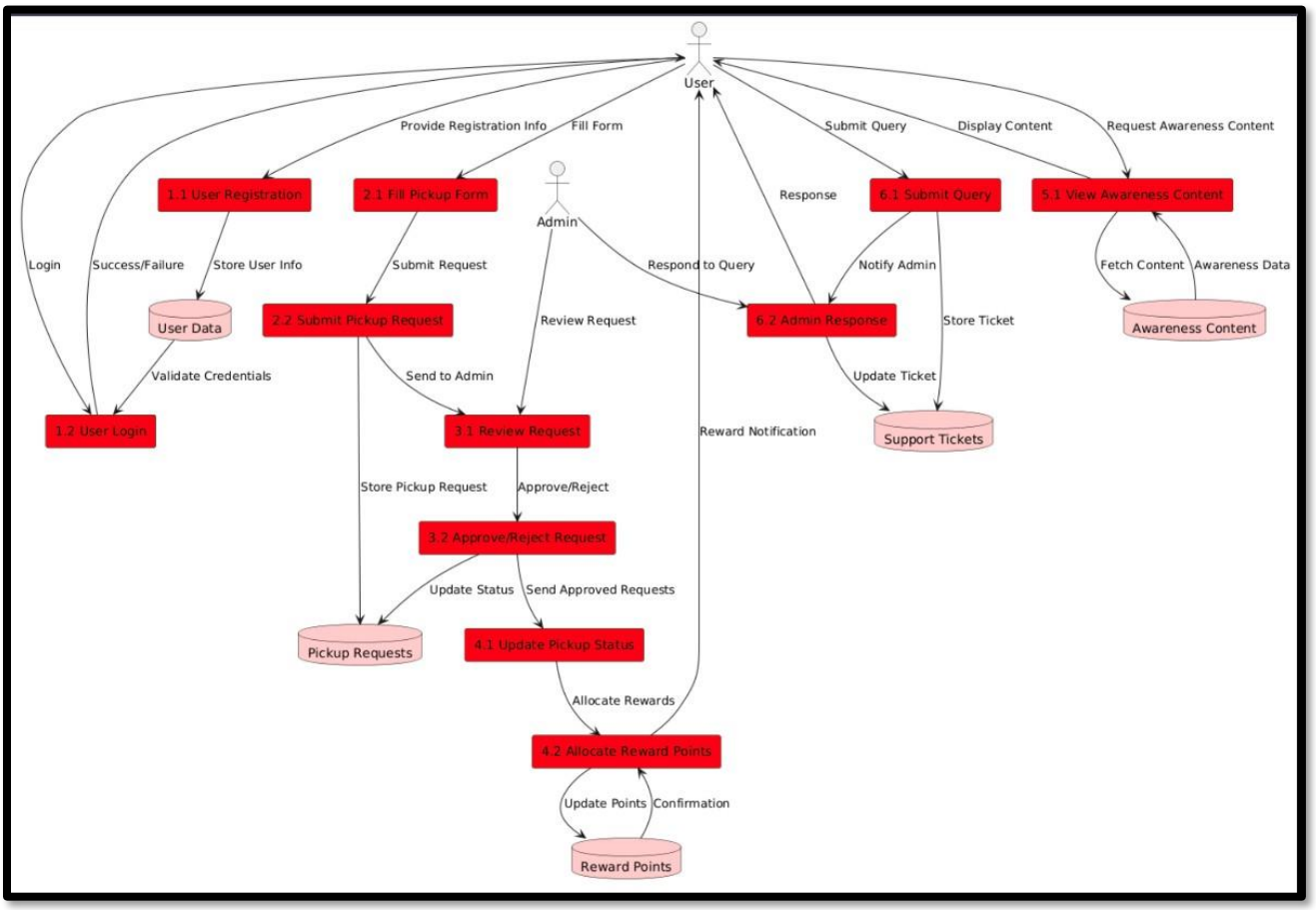


Fig. DFD Level 2

4.2 Use case diagram

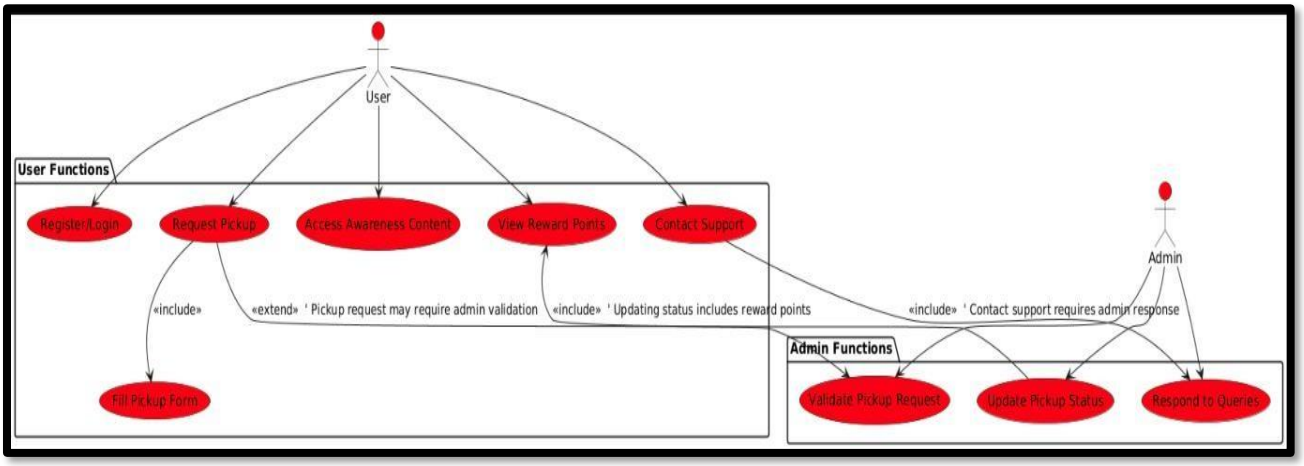


Fig. Use case Diagram

4.3 Sequence diagram

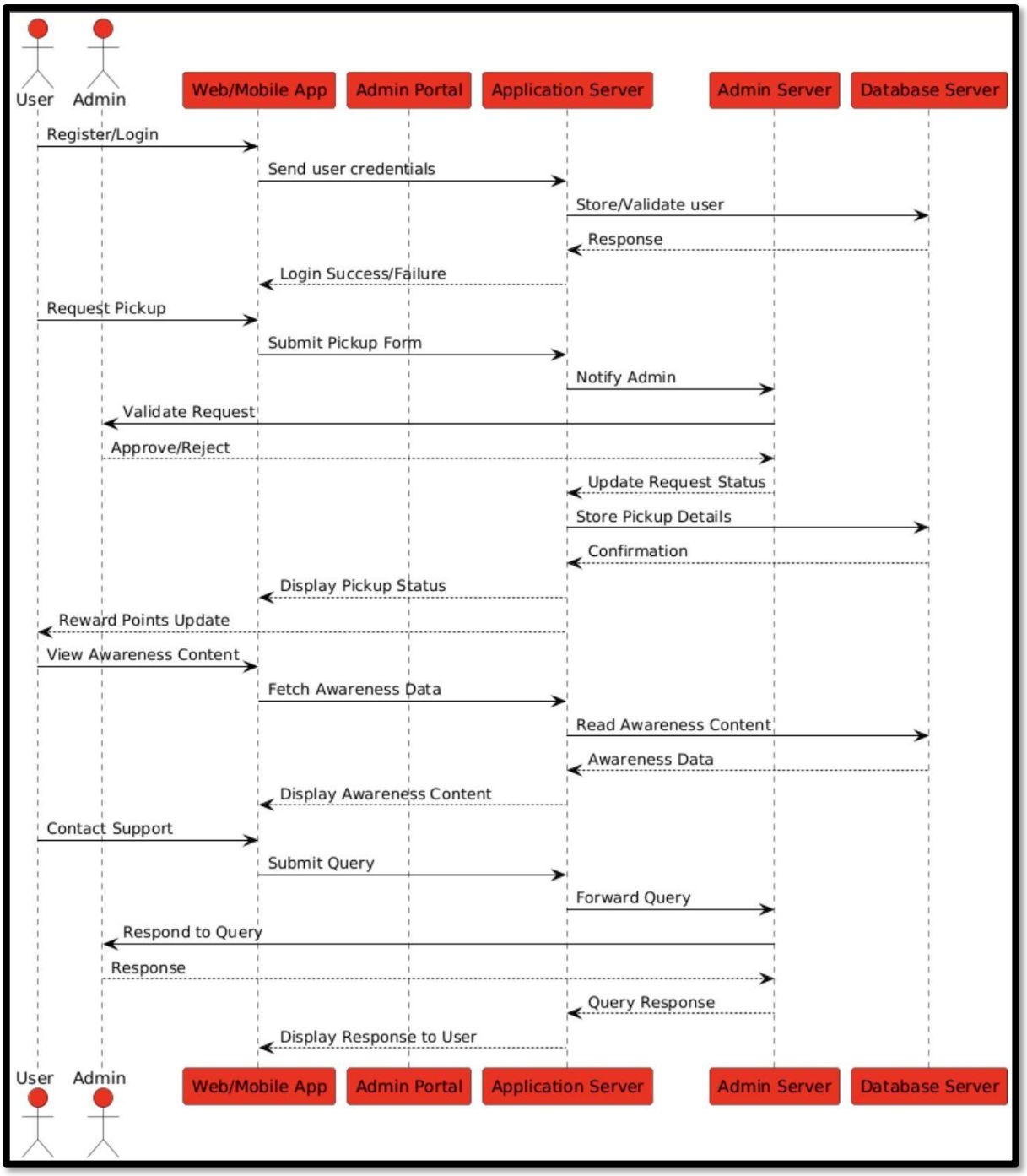


Fig . Sequence diagram

4.4 Activity diagram

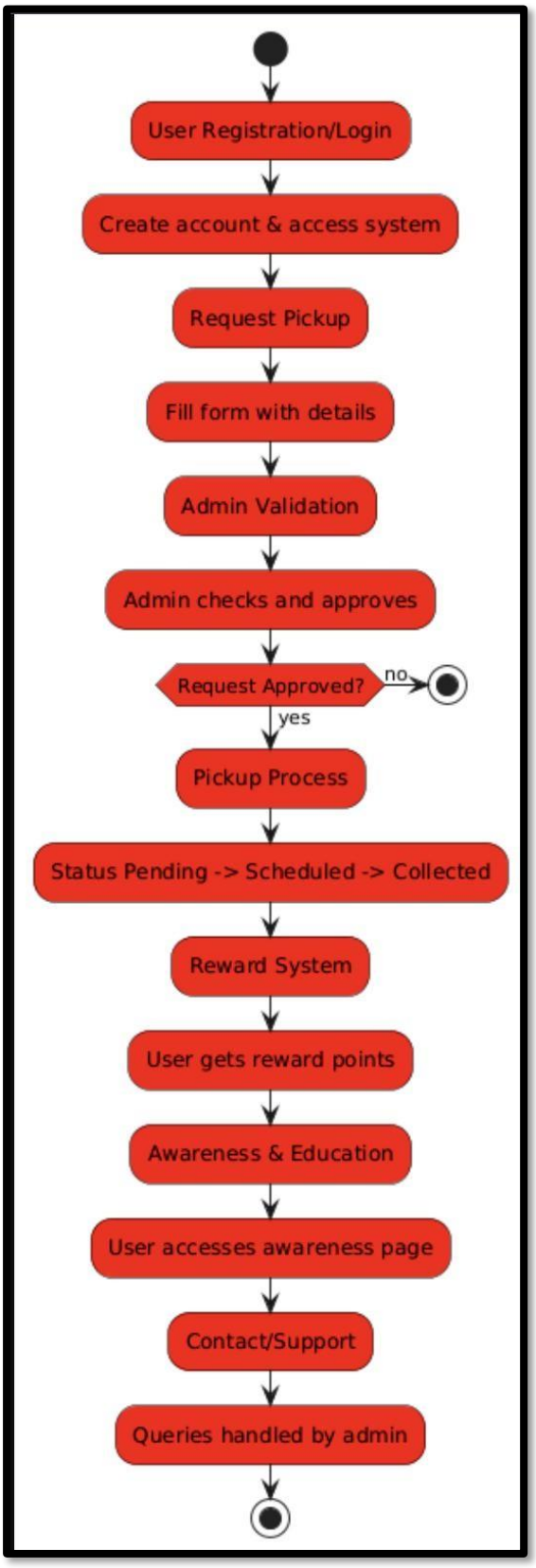


Fig. Activity diagram

4.5 Deployment diagram

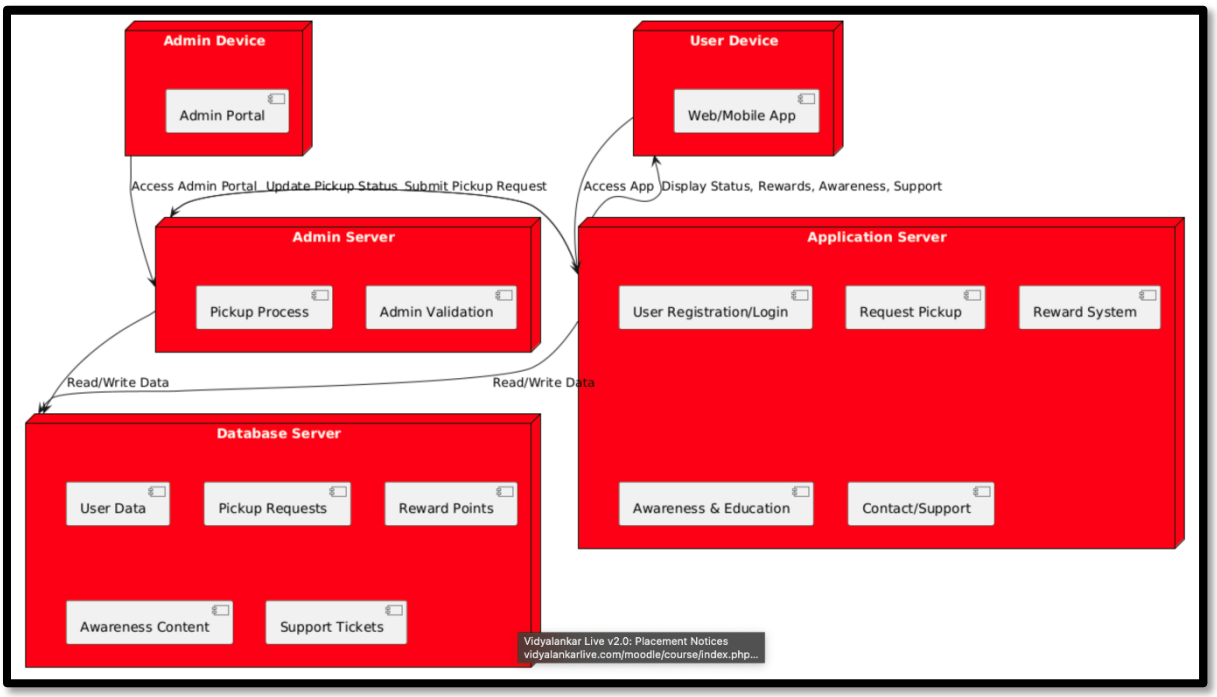


Fig. Deployment diagram

4.6 WBS

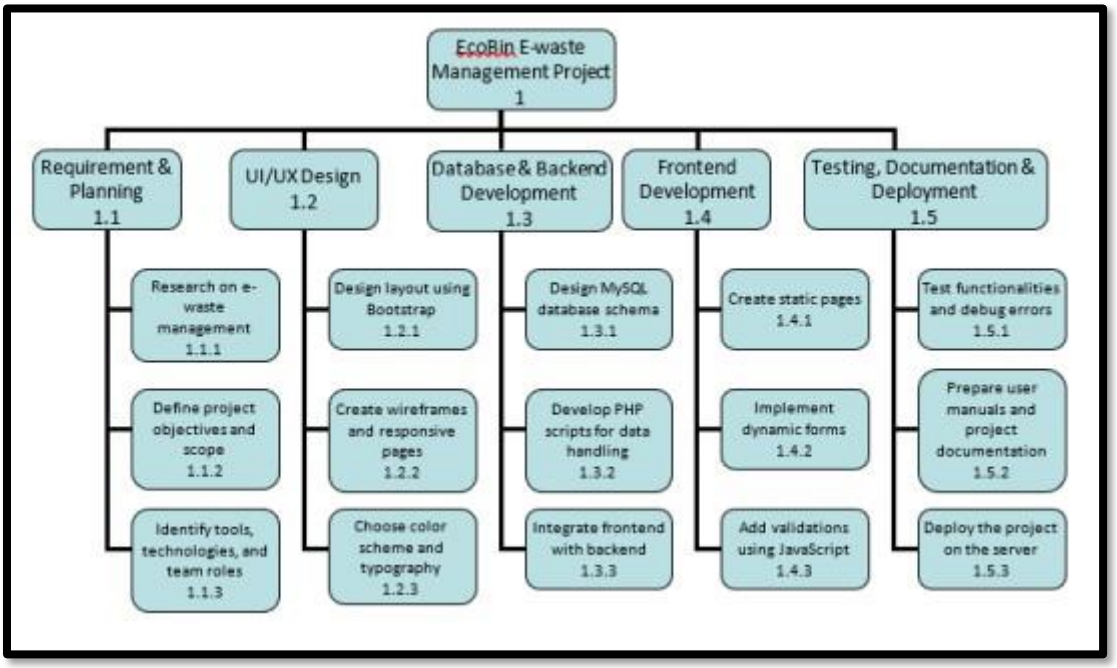


Fig. WBS

4.7 Gantt chart

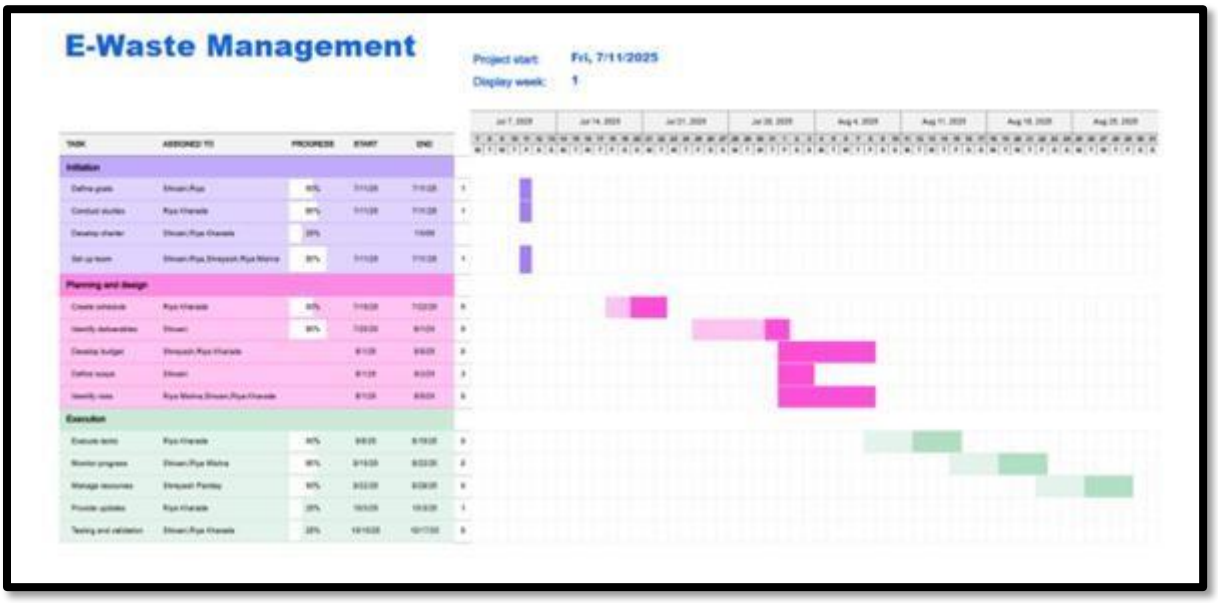


Fig. Gantt chart

4.8 RMMM Plan

Risk ID	**Risk 1**	**Type**	**R1 – User Input Error**
Probability	**High**	**Impact**	**Medium**
Risk Description	Users may enter incorrect details (type, quantity, or condition), leading to logistical issues.		
Mitigation plan	Use dropdowns, predefined options, image upload, and AI-based waste categorization.		
Monitoring plan	Track error reports from delivery partners; audit random submissions weekly.		
Management plan	Add a correction step before confirmation; allow support team to verify user input if needed.		

4.8.1 user input error

Risk ID	Risk 2	Type	R2 – Operational Delay
Probability	Medium	Impact	High
Risk Description	Delivery partners may not collect waste on time, causing user dissatisfaction.		
Mitigation plan	Integrate real-time tracking, set SLAs, and provide pickup time slots.		
Monitoring plan	Monitor pickup timestamps vs. schedule; send alerts for late pickups		
Management plan	Escalate repeated delays; reroute to alternate partner if needed.		

4.8.2 operational delay

Risk ID	Risk 3	Type	R3 – Handling Error
Probability	Medium	Impact	High
Risk Description	E-waste may be damaged or cause hazard due to poor handling.		
Mitigation plan	Train delivery partners on e-waste safety; provide handling guidelines.		
Monitoring plan	Collect delivery partner feedback; spot check handling via audit.		
Management plan	Penalize non-compliance; update training periodically.		

4.8.3 handling error

Risk ID	Risk 4	Type	R4 – Cybersecurity Risk
Probability	**Medium**	**Impact**	**High**
Risk Description	User data may be leaked or hacked if not secured properly.		
Mitigation plan	Encrypt all sensitive data; use secure authentication protocols.		
Monitoring plan	Use intrusion detection systems (IDS) and log user access.		
Management	Have a data breach response plan; notify affected users		

4.8.4 cybersecurity risk

Risk ID	Risk 5	Type	R5 – Compliance Risk
Probability	**Low**	**Impact**	**High**
Risk Description	E-waste may end up with informal/unlicensed recyclers.		
Mitigation plan	Work only with certified formal sector recyclers; verify licenses.		
Monitoring plan	Maintain chain-of-custody logs; verify recycler entries.		
Management plan	Blacklist violators; conduct quarterly compliance audits.		

4.8.5 compliance risk

Risk ID	Risk 6	Type	R6 – Technical Failure
Probability	**Low**	**Impact**	**High**
Risk Description	Users or delivery partners may be unable to use the system during downtime.		
Mitigation plan	Use cloud infrastructure with autoscaling and high availability.		
Monitoring plan	Implement uptime monitoring tools (e.g., Pingdom).		
Management plan	**Maintain emergency recovery scripts; notify users via fallback channels.**		

4.8.6 technical failure

Risk ID	Risk 7	Type	R7 – Adoption Risk
Probability	**Medium**	**Impact**	**Medium**
Risk Description	Users may not continue using the app due to lack of interest or motivation.		
Mitigation plan	Introduce gamification, rewards, and eco-points system.		
Monitoring plan	Track retention metrics; analyze churn rate.		
Management plan	Conduct user surveys; roll out engagement campaigns.		

4.8.7 adoption risk

Risk ID	Risk 8	Type	R8 – Regulatory Risk
Probability	**Low**	**Impact**	**High**
Risk Description	Platform may unintentionally violate e-waste disposal regulations.		
Mitigation plan	Consult legal experts; review compliance with e-waste rules regularly.		
Monitoring plan	Keep a log of policy changes and legal consultations.		
Management plan	Update platform features and documentation to stay compliant.		

4.8.8 regulatory risk

Risk ID	Risk 10	Type	R10 – System Abuse
Probability	**Medium**	**Impact**	**Medium**
Risk Description	Fake or spam orders may waste delivery partner time and system resources.		
Mitigation plan	Implement OTP verification and user authentication.		
Monitoring plan	Detect suspicious activity via order frequency analysis.		
Management plan	Block/report abusive users; limit order frequency.		

4.8.9 system abuse

4.9 Test cases

TEST TITLE	PRIORITY	TEST CASE ID	TEST NUMBER	TEST DATE
User Registration	High	REG-001 to REG-006	001–006	19-07-2025

TEST DESCRIPTION	TEST DESIGNED BY	TEST EXECUTED BY	EXECUTION DATE
Validate all fields in User Registration form including password, email, phone number, duplicate email, and terms checkbox.	Riya Sunil Kharade	Self	19-07-2025

TEST DESCRIPTION	TEST DEPENDENCIES	TEST CONDITIONS	TEST CONTROL
Validate all fields in User Registration form including password, email, phone number, duplicate email, and terms checkbox.	Registration page must be functional.	User attempts various valid/invalid inputs.	Manual

STEP ID	STEP DESCRIPTION	TEST DATE	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
1	Submit empty registration form	19-07-2025	Error message: "All fields are required"	Error shown	Pass	Validation works
2	Enter password pass12 (less than 8 chars)	19-07-2025	Error: "Password must be at least 8 characters"	Error shown	Pass	Password validation correct
3	Enter password pass1234 (8 chars)	19-07-2025	Password accepted	Accepted	Pass	Valid password works
4	Enter invalid email riya@123	19-07-2025	Error: "Invalid email format"	Error shown	Pass	Email regex works
5	Enter valid email riya@gmail.com	19-07-2025	Email accepted	Accepted	Pass	Correct email accepted
6	Enter phone number 12345 (less than 10 digits)	19-07-2025	Error: "Phone number must be 10 digits"	Error shown	Pass	Phone validation works
7	Enter phone number 9876543210 (10 digits)	19-07-2025	Phone number accepted	Accepted	Pass	Valid number works
8	Register with already existing email riya@gmail.com	19-07-2025	Error: "Email already exists"	Error shown	Pass	Duplicate email not allowed

9	Do not check Terms & Conditions and submit	19-07-2025	Error: "Please accept Terms & Conditions"	Error shown	Pass	Checkbox validation works
10	Check Terms & Conditions and submit valid form	19-07-2025	Registration successful	Success message shown	Pass	Registration complete

4.9.1 user registration

TEST TITLE	PRIORITY	TEST CASE ID	TEST NUMBER	TEST DATE
User Login	High	LOGIN-001 to LOGIN-005	007–011	25/07/2025
TEST DESCRIPTION		TEST DESIGNED BY	TEST EXECUTED BY	EXECUTION DATE
Verify login functionality with valid and invalid username-password combinations.		Shivani Choudhari	QA Team	25/07/2025

TEST DESCRIPTION	TEST DEPENDENCIES	TEST CONDITIONS	TEST CONTROL
Verify login functionality with valid and invalid username-password combinations.	User must already be registered.	Username and Password fields tested.	Manual

STEP ID	STEP DESCRIPTION	TEST DATE	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
7	Username valid, Password valid (TT)	25/07/2025	Login successful, redirect to dashboard	Success	Pass	Normal login
8	Username valid, Password invalid (TF)	25/07/2025	Error: "Invalid credentials"	Error shown	Pass	Wrong password
9	Username valid, Password invalid (TF)	25/07/2025	Error: "User not found"	Error shown	Pass	Wrong username
10	Username invalid, Password invalid (FF)	25/07/2025	Error: "Invalid credentials"	Error shown	Pass	Both wrong
11	Username empty, Password	25/07/2025	Error: "Username and Password required"	Error shown	Pass	Empty fields

	empty					

4.9.2 user login

TEST TITLE	PRIORITY	TEST CASE ID	TEST NUMBER	TEST DATE
Request Pickup	High	PICKUP-001 to PICKUP-010	012–021	31/06/2025
TEST DESCRIPTION		TEST DESIGNED BY	TEST EXECUTED BY	EXECUTION DATE
Verify the e-waste pickup request form functionality.		Shreyash Pandey	QA Team	31/06/2025

TEST DESCRIPTION	TEST DEPENDENCIES	TEST CONDITIONS	TEST CONTROL
Verify the e-waste pickup request form functionality.	User must be logged in. Pickup form page must be accessible.	All form fields (Item Name, Quantity, Address, Contact, Image) tested.	Manual

STEP ID	STEP DESCRIPTION	TEST DATE	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
12	Submit empty form	31/06/2025	Error: "Please fill all mandatory fields"	Error shown	Pass	Mandatory field validation
13	Enter item name only, leave other fields empty	31/06/2025	Error: "Please fill all mandatory fields"	Error shown	Pass	Partial submission blocked
14	Enter invalid contact number (less than 10 digits)	31/06/2025	Error: "Invalid phone number"	Error shown	Pass	Phone validation works
15	Enter invalid email (if email field exists)	31/06/2025	Error: "Invalid email format"	Error shown	Pass	Email validation works
16	Enter valid details, but skip image upload	31/06/2025	Form submits successfully (image optional)	Success	Pass	Email validation works
17	Enter all valid details including image	31/06/2025	Request submitted successfully, confirmation message shown	Success	Pass	Normal flow

18	Enter very large text in item description	31/06/2025	Form should handle or show error if exceeds limit	Success/Error	Pass	Field length validation
21	Enter future pickup date	31/06/2025	Request accepted, scheduled for chosen date	Success	Pass	Date selection works
21	Enter special characters in address	31/06/2025	Form accepts valid special characters	Success	Pass	Address input validated

4.9.3 request pickup

TEST TITLE	PRIORITY	TEST CASE ID	TEST NUMBER	TEST DATE
Admin Validation	High	ADMIN-001 to ADMIN-010	022–031	08/08/2025

TEST DESCRIPTION	TEST DESIGNED BY	TEST EXECUTED BY	EXECUTION DATE
Verify that admin can view, validate, and approve/reject pickup requests.	Riya Mishra	QA Team	08/08/2025

TEST DESCRIPTION	TEST DEPENDENCIES	TEST CONDITIONS	TEST CONTROL
Verify that admin can view, validate, and approve/reject pickup requests.	User must have submitted pickup requests. Admin must be logged in.	Pickup request data available, admin dashboard accessible.	Manual

STEP ID	STEP DESCRIPTION	TEST DATE	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
22	Admin logs in with correct credentials	08/08/2025	Access granted to dashboard	Success	Pass	Normal login flow
23	Admin logs in with incorrect password	08/08/2025	Error: "Invalid username or password"	Error shown	Pass	Authentication check

24	Admin views all pending pickup requests	08/08/2025	All pending requests displayed	Success	Pass	Dashboard listing works
25	Admin selects a request and approves it	08/08/2025	Status updated to "Scheduled"	Success	Pass	Approval flow works
26	Admin selects a request and rejects it	08/08/2025	Status updated to " Rejected"	Success	Pass	Rejection flow works
27	Admin updates pickup status to "Collected"	08/08/2025	Status updated successfully	Success	Pass	Status update works
28	Admin tries to approve a non-existent request	08/08/2025	Error: "Request not found"	Error shown	Pass	Validation check
29	Admin searches pickup requests by user	08/08/2025	Relevant requests filtered and displayed	Success	Pass	Search functionality
30	Admin filters requests by status (Pending, Scheduled, Collected)	08/08/2025	Filtered list displayed correctly	Success	Pass	Filter functionality
31	Admin sends confirmation notification to user after approval	08/08/2025	User receives email/SMS confirmation	Success	Pass	Notification system works

4.9.4 admin validation

TEST TITLE	PRIORITY	TEST CASE ID	TEST NUMBER	TEST DATE
Pickup Process & Reward System	High	PICKUP-STATUS-001 to PICKUP-STATUS-010	032–041	15/08/2025

TEST DESCRIPTION	TEST DESIGNED BY	TEST EXECUTED BY	EXECUTION DATE

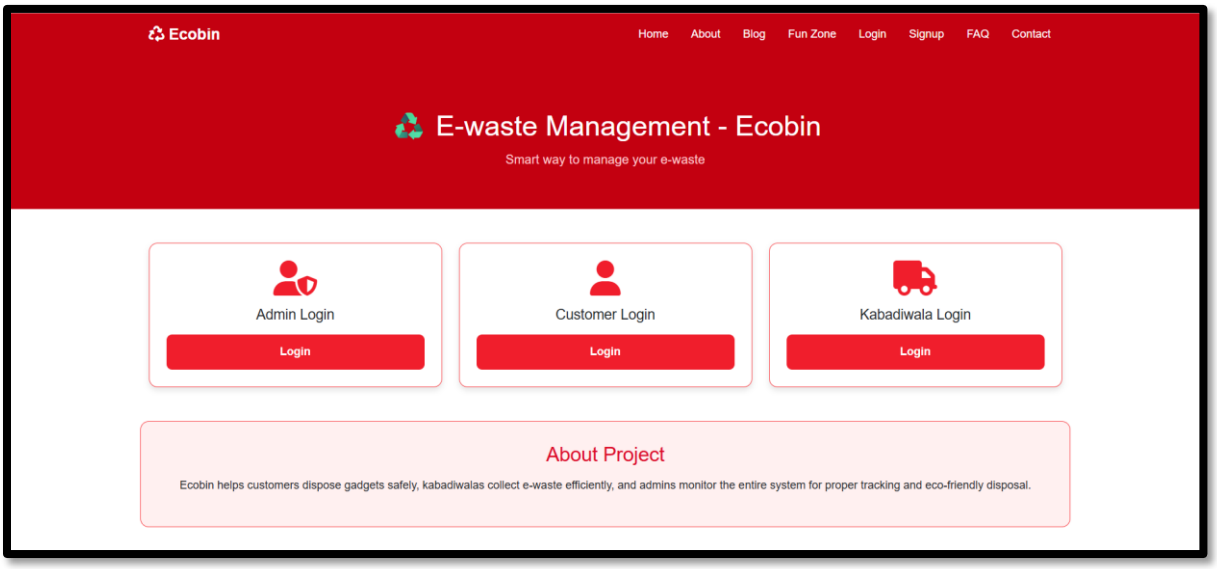
Verify that pickup requests move through proper status and reward points are assigned on completion.	Riya Sunil Kharade	Self	15/08/2025

TEST DESCRIPTION	TEST DEPENDENCIES	TEST CONDITIONS	TEST CONTROL
Verify that pickup requests move through proper status and reward points are assigned on completion.	Admin must approve pickup requests. User must be registered and logged in.	Pickup request exists in the system.	Manual

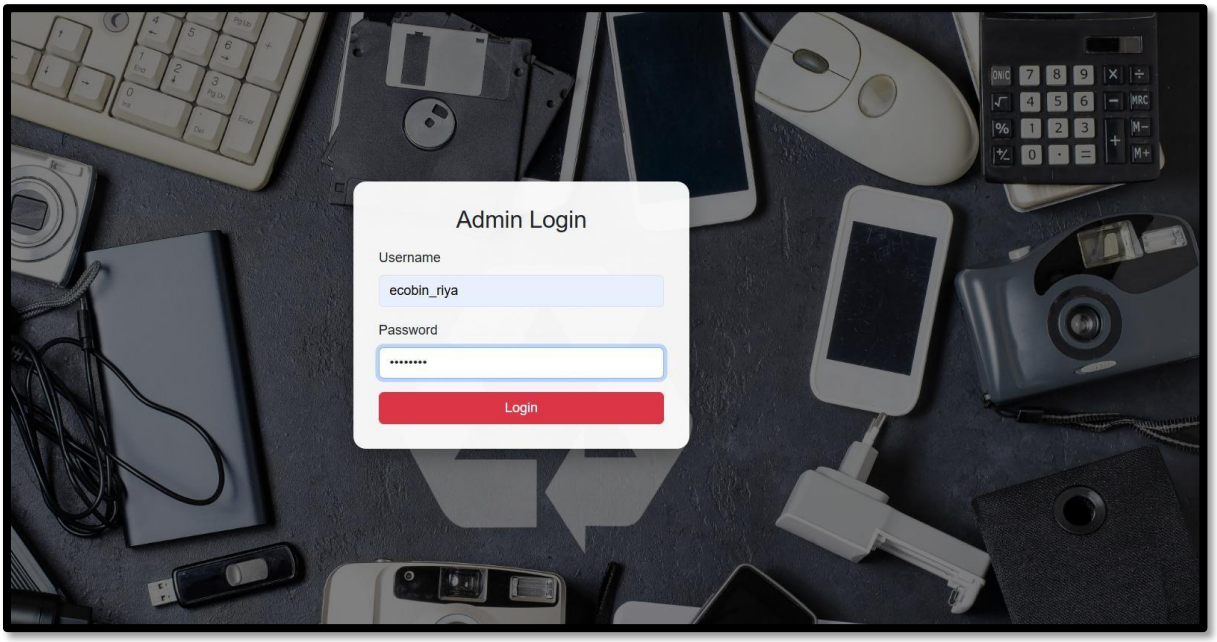
STEP ID	STEP DESCRIPTION	TEST DATE	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
32	User submits a pickup request	15/08/2025	Request created with status "Pending"	Success	Pass	Initial submission
33	Admin approves request	15/08/2025	Status changes from "Pending" to "Scheduled"	Success	Pass	Approval flow verified
34	Admin schedules pickup date/time	15/08/2025	Status shows "Scheduled" with date/time	Success	Pass	Scheduling verified
35	Pickup personnel marks request as "Collected"	15/08/2025	Status updated to "Collected"	Success	Pass	Status update works
36	System adds reward points to user account after collection	15/08/2025	Reward points reflected in user dashboard	Success	Pass	Reward assignment works
37	User views pickup history with status	15/08/2025	All past pickups and statuses displayed	Success	Pass	History view verified
38	Attempt to mark a pickup as "Collected" before approval	15/08/2025	Error: "Cannot collect unapproved request"	Error shown	Pass	Validation check

39	Check reward points do not increase for rejected request	15/08/2025	Reward points remain same	Success	Pass	Rejection flow verified
40	User redeems reward points (if feature exists)	15/08/2025	Points deducted, benefits granted	Success	Pass	Redemption tested
41	Admin checks overall reward points report	15/08/2025	Total points for all users displayed	Success	Pass	Admin report functionality

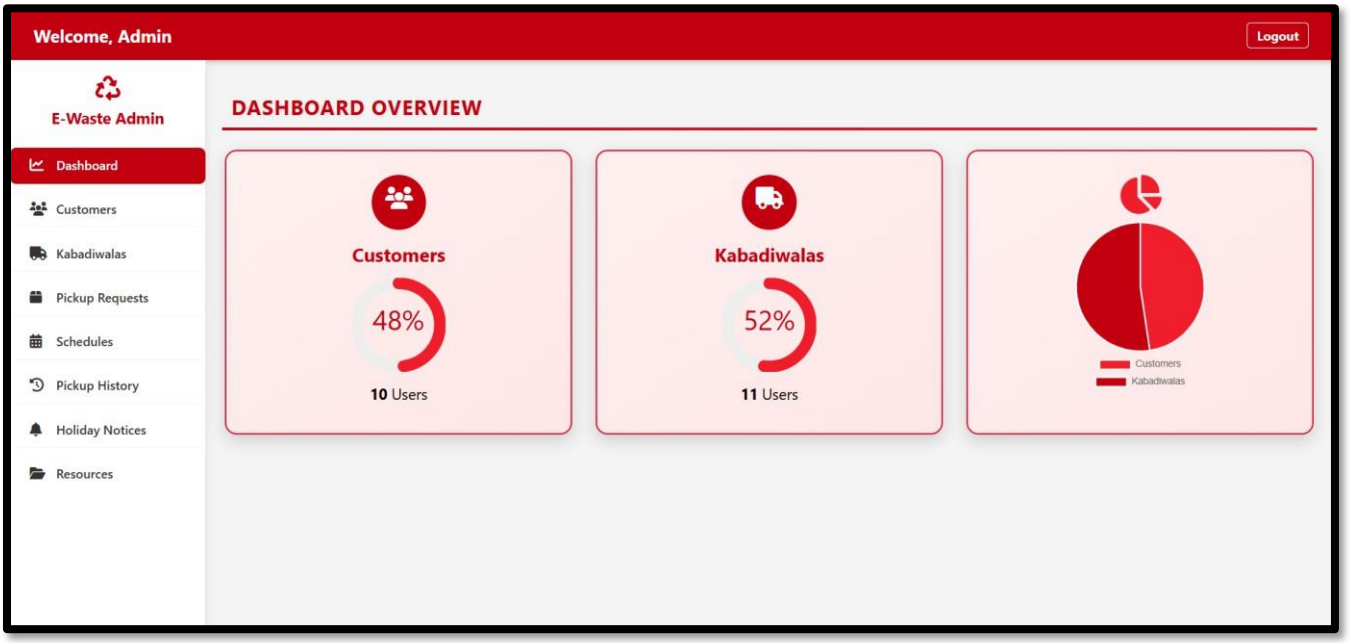
5. Results and Discussion



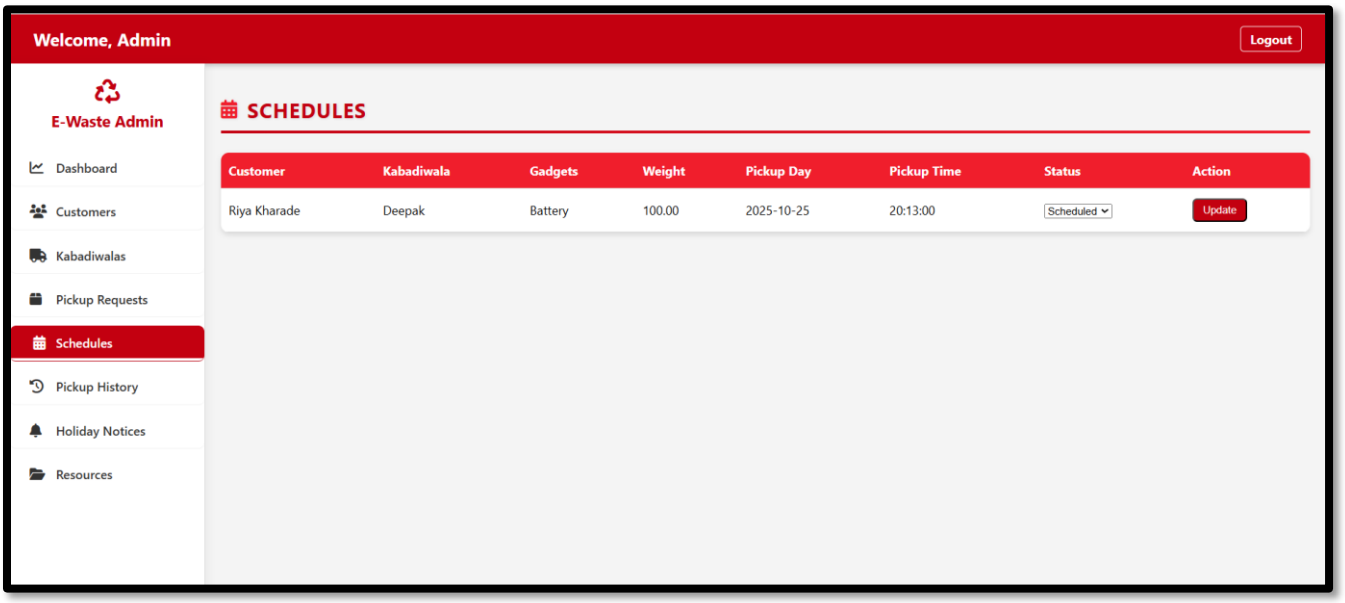
5.1 Main Webisite Homepage



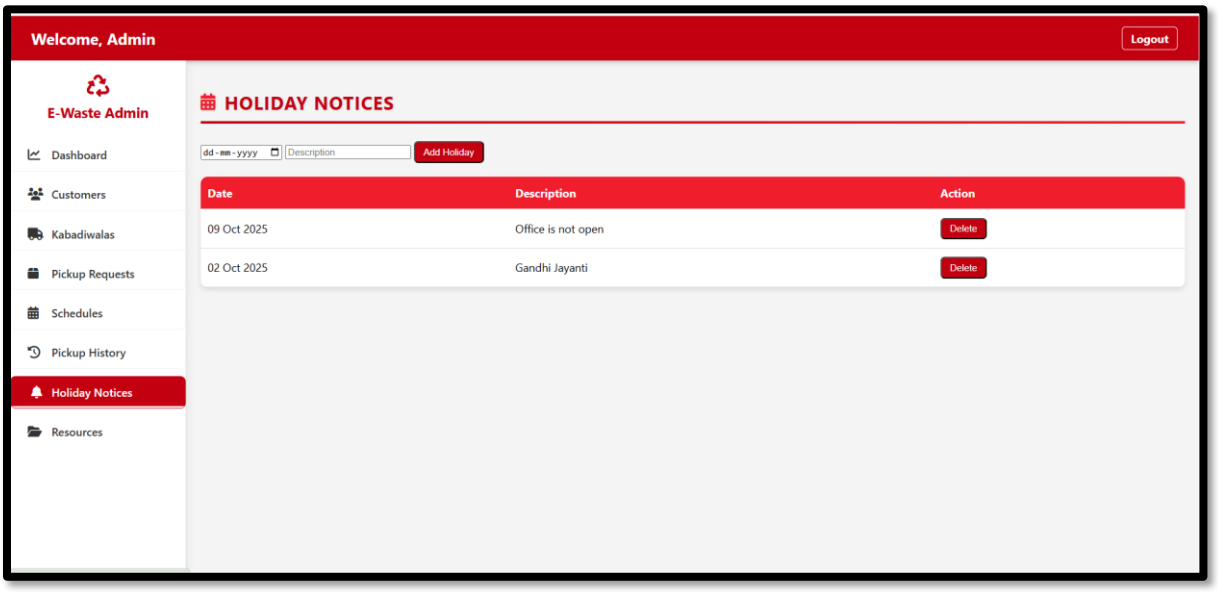
5.2 Admin Login



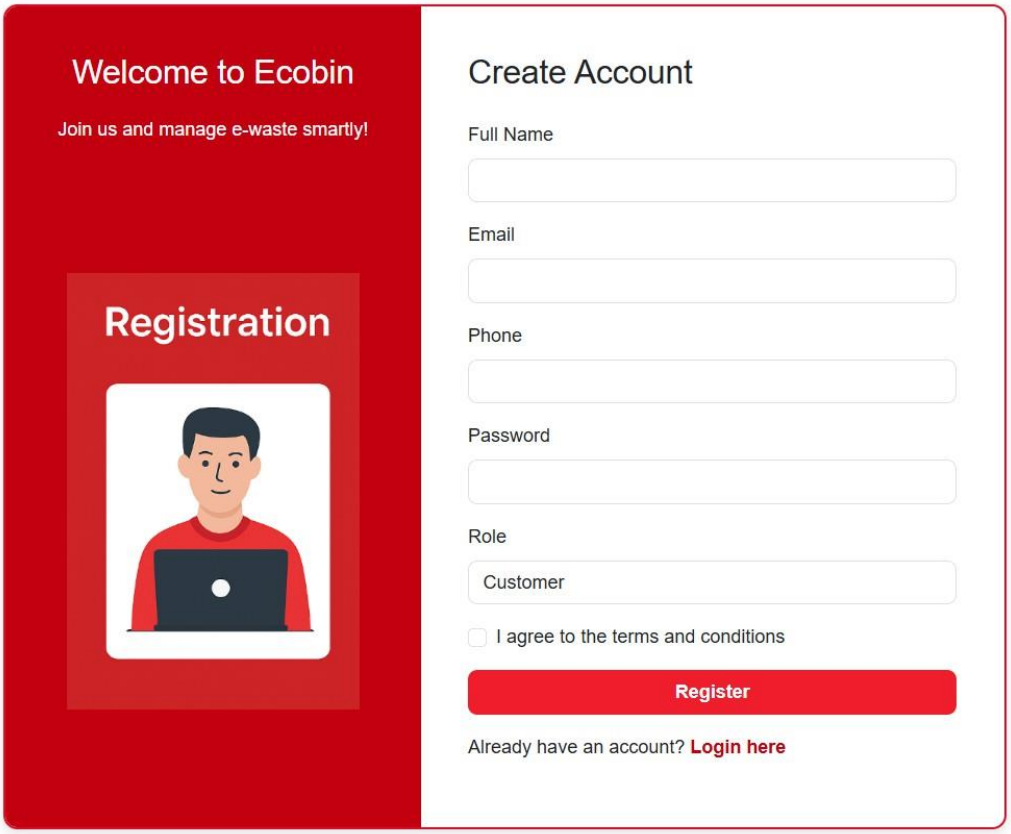
5.3 Admin Dashboard



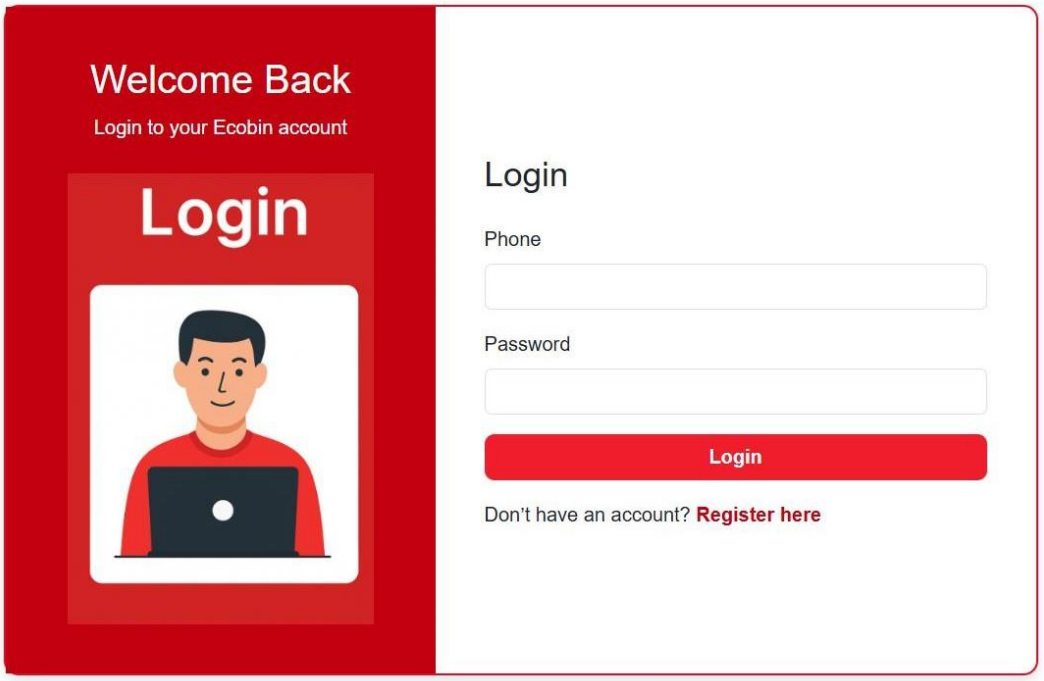
5.4 Admin Schedules Page



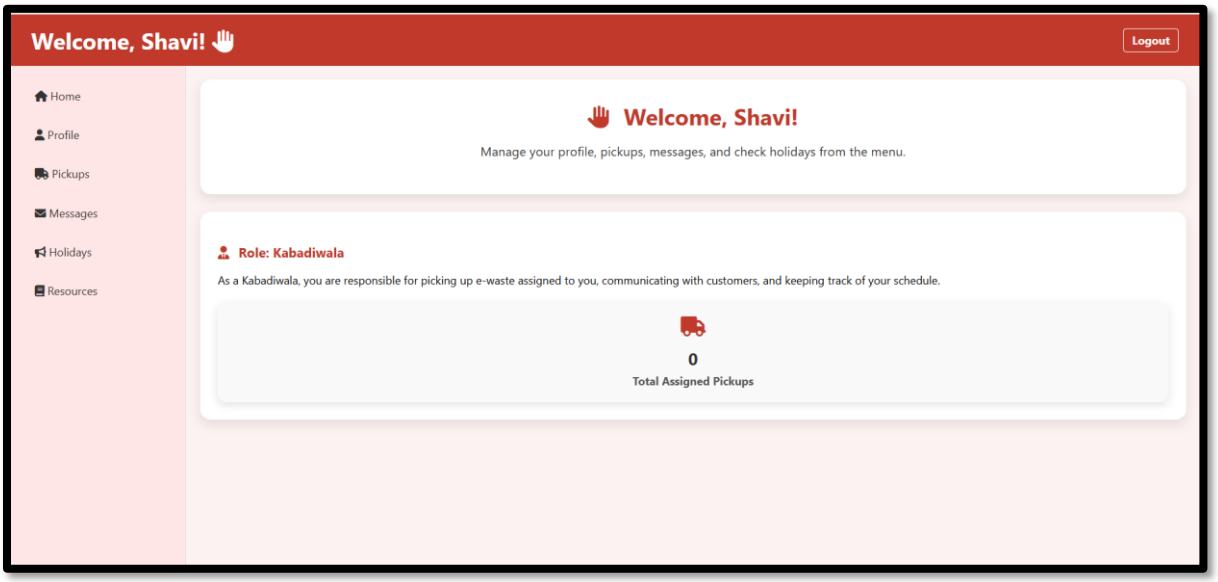
5.5 Holiday Notices Form



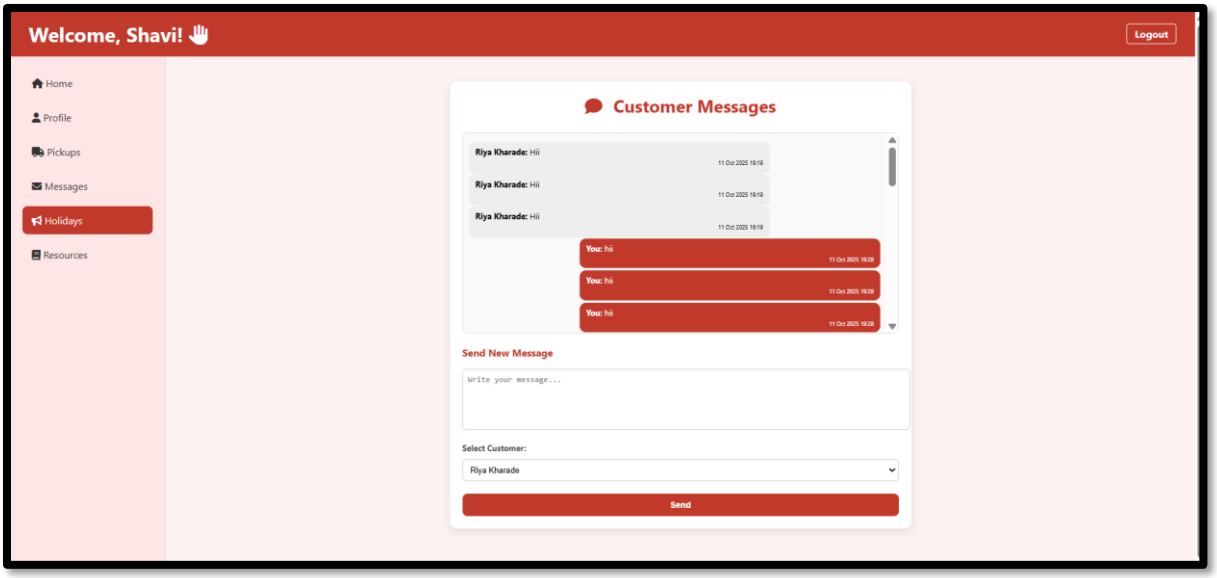
5.6 Registration For Customer and Kabadiwala



5.7 Login For Customer and Kabadiwala

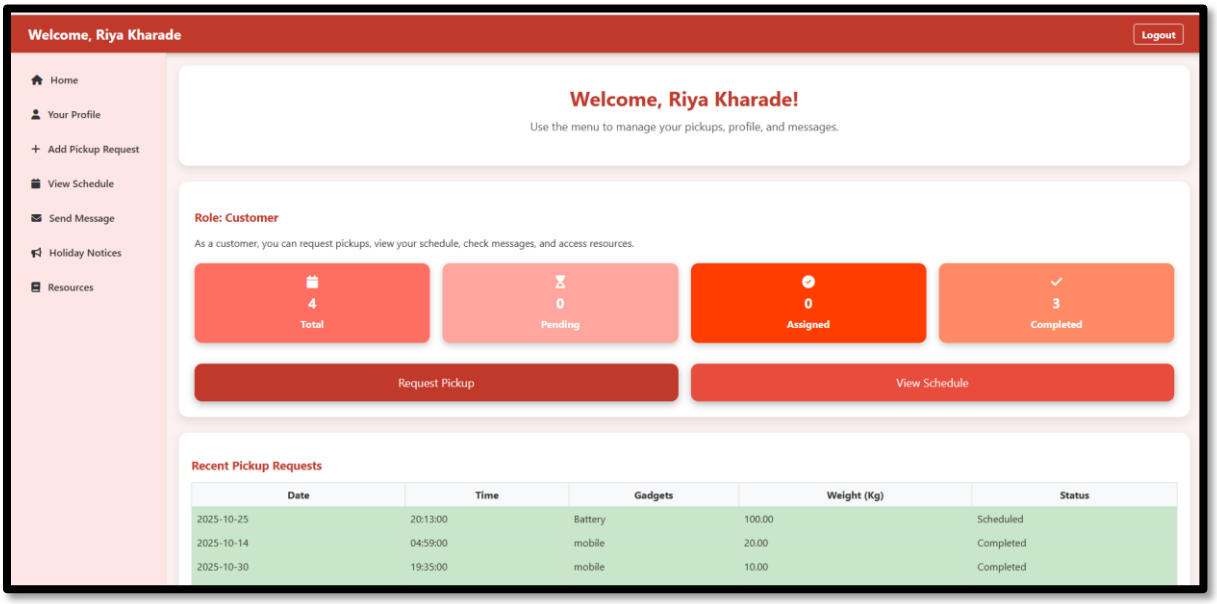


5.8 Kabadiwala Dashboard

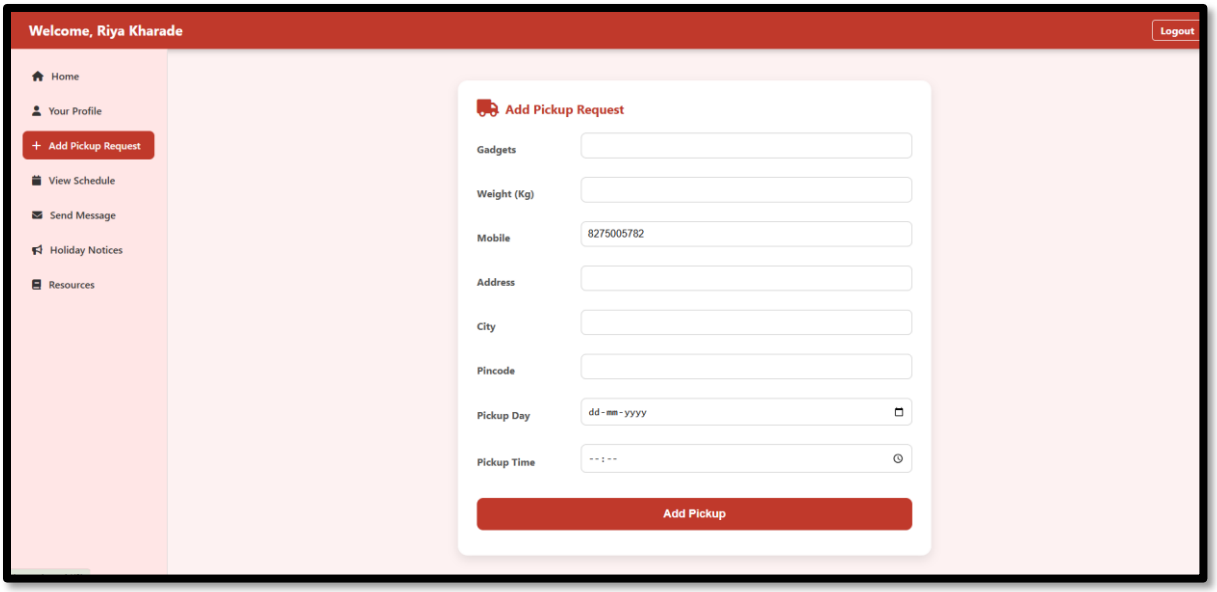


5.9 Kabadiwala Messages System

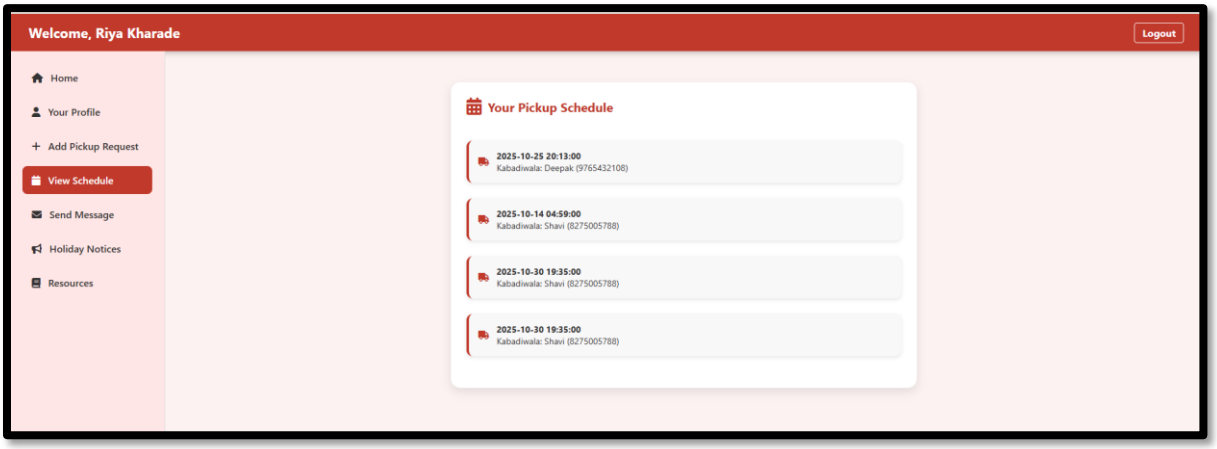
5.10 Footer of Ecobin Website



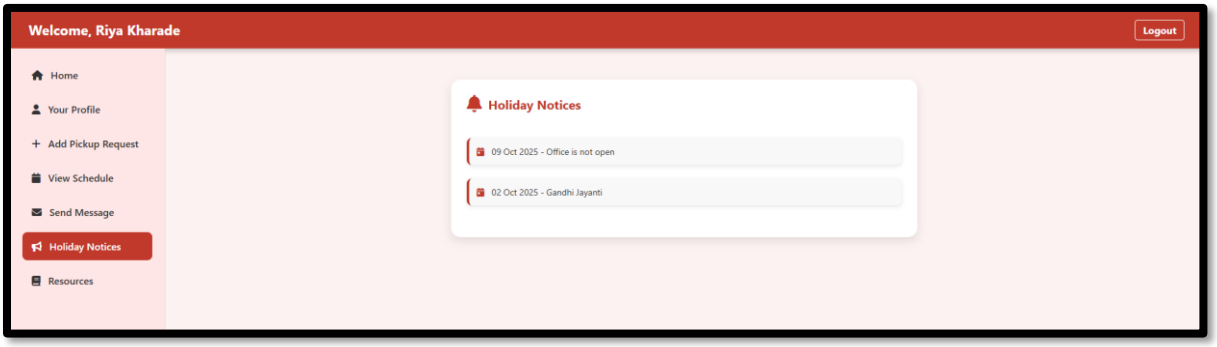
5.10 Customer Dashboard



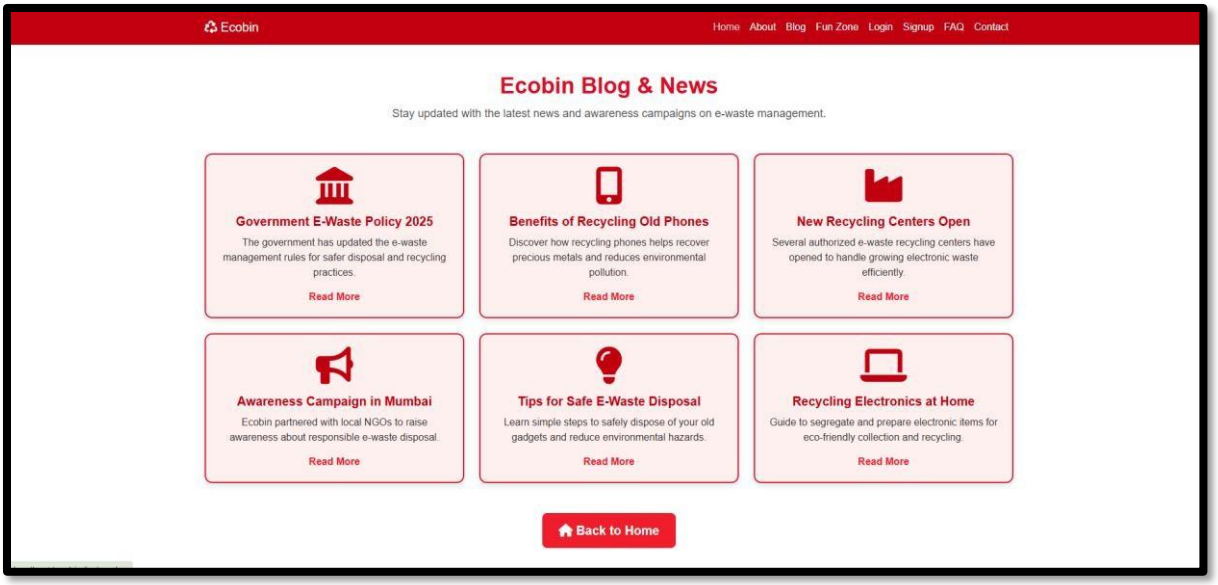
5.11 Customer Dashboard (Fill Pickup Form)



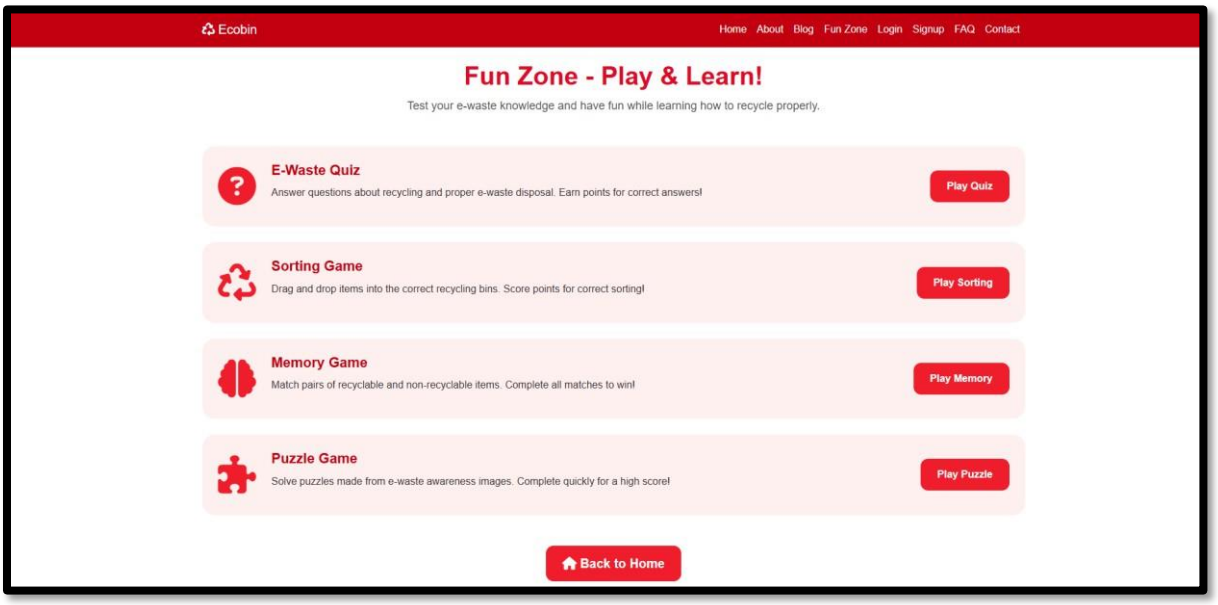
5.12 Customer Schedule View



5.13 Holiday Notice Page



5.14 Blog Section or page



5.15 Game zone

6 Conclusion and Future Scope

6.1 Conclusion

The E-Waste Management System effectively tackles the challenges of electronic waste disposal, collection, and recycling through a structured web-based platform. By connecting customers, administrators, and kabadiwalas (authorized recyclers), the system ensures efficient collection, accurate tracking, and responsible processing of e-waste. Its user-friendly interface allows easy registration, pickup requests, request tracking, and reward point management, encouraging active participation and promoting environmentally responsible behavior. The admin module maintains transparency by validating requests, assigning collection tasks, and monitoring completed pickups, while the kabadiwala module ensures timely and verified physical collection.

Additionally, the system incorporates an awareness and education module to inform users about the environmental hazards of e-waste and the importance of proper recycling practices. The integrated reward system further motivates regular participation, fostering sustainable habits among users. With secure authentication, a responsive interface, and an organized workflow, the platform successfully achieves structured e-waste management, community engagement, and environmental sustainability, demonstrating how technology can address real-world ecological challenges.

6.2 Future Scope

The E-Waste Management System holds great potential for future growth and enhancement, including integration with government programs to reward users with utility discounts, the development of a mobile app for convenient request submission and tracking, and the use of advanced analytics and AI for efficient collection and e-waste classification. The platform can also expand to involve corporate and institutional partnerships, enable direct monetary rewards or donations through payment gateways, and provide multilingual support to reach a wider audience. By combining technology, incentives, and education, the system offers a scalable and adaptable solution that promotes responsible e-waste management, environmental awareness, and a sustainable future for individuals, communities, and governments.

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